

Heart Disease MLOps Assignment

This project implements an end-to-end MLOps workflow for heart disease prediction using the UCI Heart Disease dataset. The solution covers data preparation, exploratory analysis, model training, experiment tracking, packaging, API serving, CI/CD, monitoring, and deployment.

Problem Statement

Build a machine learning classifier to predict the risk of heart disease from patient health data and prepare the solution as a cloud-ready, monitored API.

Project Structure

- `data/raw/` - original dataset files
- `data/processed/` - cleaned and prepared datasets
- `src/data/` - dataset loading and inspection scripts
- `src/features/` - cleaning, target preparation, preprocessing, and EDA scripts
- `src/models/` - training, tuning, tracking, and packaging scripts
- `src/api/` - FastAPI application and monitoring demo
- `tests/` - unit tests
- `.github/workflows/` - GitHub Actions CI pipeline
- `k8s/` - Kubernetes deployment manifests and guide
- `artifacts/` - generated plots, summaries, logs, screenshots, and reports
- `models/` - final packaged model and metadata

Dataset

This project uses the processed Cleveland subset of the UCI Heart Disease dataset.

Required dataset file:

- `data/raw/processed.cleveland.data`

Optional reference file:

- `data/raw/heart-disease.names`

Environment Setup

Install dependencies:

```
python -m pip install --upgrade pip
pip install -r requirements.txt
```

Execution Order

Run the main project workflow in this order:

```
python src\data\data_understanding.py
python src\features\clean_data.py
python src\features\prepare_target.py
python src\features\eda.py
python src\models\train_models.py
python src\models\tune_models.py
python src\models\track_experiments.py
python src\models\package_model.py
python -m pytest tests
```

MLflow Experiment Tracking

Run experiment tracking:

```
python src\models\track_experiments.py
```

Open the MLflow UI:

```
mlflow ui
```

Then open:

- <http://127.0.0.1:5000>

FastAPI Inference Service

Run the API locally:

```
uvicorn src.api.main:app --reload
```

Local API endpoints:

- GET /
- GET /health
- GET /metrics
- POST /predict

Local Swagger UI:

- <http://127.0.0.1:8000/docs>

Public Deployment

The API is publicly deployed on Render.

Public deployed URL:

- <https://heart-disease-api-lo3h.onrender.com>

Public endpoints:

- Swagger UI: <https://heart-disease-api-lo3h.onrender.com/docs>

- Health endpoint: `https://heart-disease-api-lo3h.onrender.com/health`
- Predict endpoint:
`https://heart-disease-api-lo3h.onrender.com/predict`

API Testing

To test the API locally or on the public deployment:

1. Open the Swagger UI.
2. Expand `POST /predict`.
3. Click `Try it out`.
4. Use the sample JSON request.
5. Click `Execute`.
6. Confirm that the response returns `prediction` and `confidence`.

Docker and Containerization

The project includes containerization support through the repository configuration.

The intended setup packages:

- Python runtime
- project dependencies
- FastAPI application
- saved model artifacts
- startup configuration

During development, full local container execution could not be completed on the office laptop because Docker-based tooling required additional platform components and administrator-level installation support. Because of this setup limitation, the container configuration was prepared in the repository, while the final working deployment was completed through Render using the cloud build process.

CI/CD

GitHub Actions workflow file:

- `.github/workflows/ci.yml`

The CI pipeline performs:

- dependency installation
- linting with `flake8`
- unit testing with `pytest`

- model training and packaging validation

Kubernetes Deployment

Prepared deployment files:

- k8s/heart-disease-deployment.yaml
- k8s/heart-disease-service.yaml
- k8s/deployment_guide.md

These files are included as deployment-ready manifests for Kubernetes-based deployment.

Key Results

- Final dataset size: 303 rows
- Final target: binary target
- Best model: logistic_regression_tuned
- Test ROC-AUC: 0.966
- Test Recall: 0.929
- Test F1-score: 0.881

Important Artifacts

- EDA findings: artifacts/eda/eda_findings.md
- Model comparison: artifacts/model_training/model_comparison.csv
- Tuned model summary: artifacts/model_tuning/tuned_model_summary.md
- Experiment summary:
artifacts/experiment_tracking/experiment_summary.md
- Packaging summary: artifacts/model_packaging/packaging_summary.md
- Monitoring summary: artifacts/monitoring/monitoring_summary.md
- Screenshots: artifacts/screenshots/
- Final report: artifacts/report/

Repository Link

GitHub repository:

- <https://github.com/2024ac05094-Riyaz-Bits/heart-disease-mlops-assignment>